

Mathumitha Rajan

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OBJECTIVE: M.S. Electrical and Computer Engineering student seeking RTL Design, Design Verification, or Validation internship/Full time opportunities, with hands-on experience in SystemVerilog, UVM, computer architecture, and hardware validation.

EDUCATION

M.S., Electrical and Computer Engineering | Portland State University, Portland, Oregon (Expected 06/27)

Coursework: Microprocessor System Design, Computer Architecture, Post-Silicon Validation, Pre-Silicon Validation(UVM), ASIC Modeling and Synthesis. **GPA: 3.85/4.0**

Upcoming Courses: Advanced Computer Architecture, Assertion Based Verification and Formal Verification.

B.Tech, Electronics and Communication Engineering | Amrita Vishwa Vidyapeetham, Coimbatore, India (2024)

TECHNICAL SKILLS

Languages –SystemVerilog, Python, C/ C++.

RTL Desing & Verification - RTL Design, UVM, Constrained-Random Verification, SystemVerilog Assertions (SVA), Functional & Code Coverage, Logic Equivalence Checking (LEC)

Computer Architecture - Pipelined Processors, Cache Coherence (MESI/MSI), Instruction Set Architectures, DDR Controllers, Memory Hierarchy, Branch Prediction

Tools & Protocols – Siemens QuestaSim/ModelSim, Synopsys DC, Synopsys Primetime, Vivado Siemens QuestaSim/ModelSim, Synopsys DC, Synopsys PrimeTime, Vivado, GDB; AXI, APB

ASIC Design Flows – Synthesis, Static Timing Analysis (STA), Clock Domain Crossing (CDC), Scan/ DFT fundamentals

Automation – Tcl, Bash/Shell scripting, Makefile automation.

TECHNICAL PROJECTS

5-Stage Pipelined MIPS CPU RTL Design & UVM Testbench [SystemVerilog | UVM]

- Implemented a 5-stage pipelined MIPS CPU supporting forwarding, hazard detection, and branch recovery.
- Developed a UVM verification environment with scoreboards and a SystemVerilog reference model to validate ALU and pipeline functionality.
- Achieved >90% functional coverage** through directed and constrained-random testing while debugging pipeline corner cases.

Dual-Core MESI Cache-Coherence Subsystem [SystemVerilog]

- Designed a **2-core, 4-state MESI cache-coherence subsystem** with private split L1 caches (**4-way I-cache, 8-way D-cache, 64B lines**), a shared snooping bus, and a shared L2, using write-back/write-allocate and true-LRU replacement.
- Implemented **cache-controller logic** for all MESI transitions and bus transactions (**BusRd / BusRdX / BusUpgr**), handling snoops, ownership transfer, and dirty-line eviction across cores.
- Verified the RTL with a **self-checking testbench of 28 directed tests** (100% passing) and characterized D-cache hit ratios from **~94% to ~0%** across sequential, strided, and conflict-heavy workloads.

Multiprocessor Random Instruction Test Generator — Post-Silicon Validation [C | x86-64 | GDB]

- Developed a seed-driven x86-64 instruction generator that emits executable machine code at runtime, producing **25+ randomized instructions** per execution.
- Implemented multi-process validation using shared memory, XADD/XCHG, and memory fences to **exercise atomic operations and memory-ordering behavior**.
- Debugged generated instruction streams using GDB, validating instruction encoding, register preservation, stack integrity, and memory accesses.

IEEE-754 Single-Precision Floating-Point Adder Design and Verification [SystemVerilog]

- Designed a **32-bit IEEE-754 floating-point adder** supporting alignment, normalization, round-to-nearest-even, and exception handling.
- Verified bit-exact against a shortreal reference model with a **100% pass rate across 40+ corner-case tests**, debugging overflow, underflow, denormals, and cancellation

PROFESSIONAL EXPERIENCE

Hardware Engineer, IIT Madras, India

June 2024 – February 2025

- Integrated FTDI FT601 SuperSpeed USB 3.0 interface with FPGA firmware, increasing host communication throughput from 480 Mbps to 3.2 Gbps for real-time ultrasound acquisition systems.
- Debugged and validated USB, PCIe, UART, and I2C interfaces during FPGA board bring-up by analyzing timing, protocol, and data-transfer issues, improving hardware integration and system reliability.

PUBLICATIONS

Mathumitha R, et al., "Low Power, High Accuracy Approximate Multiplier for Error-Resilient Image Processing Application" 2024 IEEE International Conference on Distributed Computing, VLSI, Electrical Circuits and Robotics (DISCOVER), Mangalore, India, 2024, pp. 159-163